Slides, posters, and figures

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Talks, posters, and figures carry the same research as a paper, but the reader sees them for less time and cannot reread them. This chapter gives practical rules for each.

Guidelines for slide decks

When preparing slides for a lab meeting, conference, qualifying exam, or thesis defense, follow these guidelines:

- **Always include slide numbers**: Slide numbers make feedback specific. During questions or a practice run, listeners need to refer back to a slide quickly (e.g., "On slide 14, how was that variable defined?") instead of describing it or guessing where a topic occurred.
- **Complete, record, and review practice runs**: Before any significant presentation, rehearse the complete talk out loud, record the practice run (with screen recording software, a webcam, or your phone), and watch the recording. The recording shows your pacing, timing, filler words, and transitions, and whether slide text and figures stay legible on screen.
- **One main point per slide**: Limit the cognitive load of each slide. If a slide presents several findings or a complex argument, split it across consecutive slides so the audience can follow each point.
- **Prefer visual evidence**: Audiences cannot read dense paragraphs while listening to a speaker. Use clean charts, diagrams, and annotated figures instead of text-heavy bullet lists.
- **Ensure high contrast and readability**: Use large fonts and high-contrast colors. Check that axis labels, legends, and annotations can be read from the back of a large room or on a low-resolution projector.

Presentation and poster resources

- How to tell a compelling story in scientific presentations (Kirchoff 2021)
- The science of storytelling: the David Attenborough style of scientific presentation (Ratcliff 2023)
- How to create a better research poster in less time - Mike Morrison's video introducing the #betterposter template
- Towards an "#evenbetterposter" - the MIT Biological Engineering Communication Lab's improvements to that template

Show the result in a figure

A figure often shows a pattern, an outlier or a comparison faster and more reliably than a sentence or a table can describe it. So when a result can be drawn, draw it, and use the text to point the reader to what the figure shows.

- Show the data before summarizing them. Summary statistics can hide structure that a plot of the data reveals.
- Put the key result of each section in a figure, and write the section's text around that figure.
- Make the figure show the comparison the text claims, with the groups, fitted curve or interval on the same axes.
- Say in the text what to look at in the figure (see Figures and tables).
- Use a table instead when readers need exact values to look up or compute with.

Example 0.1 (Four datasets with the same summaries)**.** Anscombe (1973) constructed four small datasets of (x, y) pairs (Wikipedia: Anscombe's quartet). All four have the same mean of x (9), the same mean of y (7.5), the same correlation (0.82 to two decimal places) and the same least-squares line, with slope 0.5. A table of these summaries cannot tell the datasets apart. A scatterplot of each can: one is roughly linear with scatter, one follows a smooth curve, one is an exact line with a single outlier, and one has all but one of its points at the same x.

Scientific figures

A clear figure lets a reader check a result without help from the text. Before you share a figure:

- Label the x-axis and y-axis clearly.
- Use panel labels when a figure has several subplots.
- Include enough tick marks and tick labels.

- Print tick labels as plain or comma-separated numbers, such as 0.001 or 10,000, not in scientific notation. A logarithmic axis is fine. Only the label format matters, so write 1, 10, 100 and 1,000, not 1e+00 through 1e+03. When the numbers are huge or tiny, rescale the units and say so in the axis title, such as "dollars (millions)".
- Use font sizes that are easy to read without dominating the plot.
- Use colorblind-friendly palettes (see ColorBrewer or iwanthue).
- Define colors, shapes, and line types in a legend.
- Minimize abbreviations in labels and legends.
- Use transparency to show overlapping data.
- Show measures of precision, such as 95% confidence intervals.

For more:

- Ten Simple Rules for Better Figures (Rougier et al. 2014)
- Tufte CSS - style HTML articles like Edward Tufte's books and handouts, with sidenotes, typography, and tight integration of graphics with text

Source

Adapted from the Professional Development and Resources chapters of the UCD-SERG lab manual. Several of the resource links began in the Benjamin-Chung Lab Manual by Jade Benjamin-Chung.

Anscombe, F. J. 1973. "Graphs in Statistical Analysis." *The American Statistician* 27 (1): 17–21. https://doi.org/10.1080/00031305.1973.10478966.

Kirchoff, Bruce. 2021. "How to Tell a Compelling Story in Scientific Presentations." *Nature* 600 (7890): S88–89. https://doi.org/10.1038/d41586-021-03603-2.

Ratcliff, William C. 2023. "The Science of Storytelling: The David Attenborough Style of Scientific Presentation." *Trends in Molecular Medicine* 29 (9): 675–77. https://doi.org/10.1016/j.molmed.2023.05.002.

Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. 2014. "Ten Simple Rules for Better Figures." *PLOS Computational Biology* 10 (9): e1003833. https://doi.org/10.1371/journal.pcbi.1003833.